Plant Leaves Super-Resolution Challenge

Technical Report

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1. Introduction

The task is the Plant Leaves Super-Resolution Challenge on Kaggle. The objective is to reconstruct high-resolution plant leaf images from degraded low-resolution inputs. Each input is a 32×32 RGB image and the expected output is a 128×128 RGB image, requiring the model to learn a $4\times$ super-resolution mapping.

Plant leaf images contain fine texture, edges, disease patterns, and structural detail that are useful for downstream agricultural analysis. Simple interpolation methods such as bicubic resizing enlarge an image but cannot reliably recover missing high-frequency information.

1.1. Dataset

The dataset was provided by the Kaggle competition and is part of the AgriVision collection. The notebook confirmed the following structure:

```
============================================================
DATASET OVERVIEW
============================================================
Training LR images:  1642
Training HR images:  1642
Test LR images:      495
Scale factor:        4x (32x32 -> 128x128)
Pixels per HR image: 128x128x3 = 49,152
```

Data loading confirmed paired tensors with the following properties:

```
Found 1642 image pairs.
Found 495 test images.
LR shape: torch.Size([3, 32, 32]),  HR shape: torch.Size([3, 128, 128])
LR range: [0.000, 0.710]
HR range: [0.000, 1.000]
```

The notable asymmetry in the LR pixel range $[0, 0.710]$ versus the HR range $[0, 1.000]$ indicates that the degradation process compresses the dynamic range in addition to reducing spatial resolution, making the reconstruction task harder than simple upscaling.

1.2. Baseline

Before training any model, bicubic interpolation was evaluated on all 1,642 training pairs:

```
==================================================
  Bicubic Baseline MAE (0-255): 18.40
  This is the score to beat with our cGAN.
```

```
==================================================
```

An MAE of 18.40 represents the performance ceiling of naive upsampling without any learned reconstruction. All subsequent results are compared against this baseline.

2. Methodology

2.1. Architecture

Generator — RRDB-Net with Channel Attention

The generator is an RRDB-Net inspired architecture built from 20 Residual-in-Residual Dense Blocks (RRDB), each containing three stacked Residual Dense Blocks (RDB). Each RDB has five densely connected convolutional layers with a growth rate of 32, followed by Squeeze-and-Excitation (SE) channel attention with a reduction ratio of 16. Residual scaling of $\beta = 0.2$ is applied at both the RDB and RRDB level.

Upsampling uses two sequential PixelShuffle stages (each 2×) for a total 4× factor. Both PixelShuffle convolutions are initialised with ICNR (Integer Convolution Nearest-neighbour Resize) to eliminate checkerboard artefacts at initialisation. All other convolutional layers use Kaiming Normal initialisation with LeakyReLU gain.

The notebook confirmed the model scale:

```
Generator -- Total params: 14,792,003 | Trainable: 14,792,003
```

Architecture summary: `Conv` → 20× `RRDB(+SE)` → `Conv` → `PixelShuffle(2x)` → `PixelShuffle(2x)` → `Conv`.

Discriminator — U-Net PatchGAN

Instead of a standard PatchGAN, a U-Net discriminator was used to provide richer per-pixel gradient signals back to the generator. The encoder downsamples from 128 → 64 → 32 → 16 → 8 pixels using strided convolutions with InstanceNorm. The decoder upsamples with skip connections from each encoder stage, producing a 128 × 128 map of per-pixel real/fake logits.

```
Discriminator -- Total params: 8,656,412
```

The discriminator was used only in Phase 2; the adversarial loss weight was kept at $\lambda_{\text{adv}} = 0.0002$ to avoid degrading the pixel accuracy achieved in Phase 1.

2.2. Preprocessing and Augmentation

- Images loaded as RGB with PIL; converted to float32 tensors in $[0, 1]$.
- Training set augmented with random horizontal flips, random vertical flips, and random 90 rotations ($k \in \{0, 1, 2, 3\}$), applied jointly to LR–HR pairs to maintain alignment.
- Validation-style MAE was computed without augmentation.
- DataLoaders used batch size 16, 2 worker processes, and pinned memory for GPU throughput.

2.3. Loss Functions

Charbonnier Loss

Charbonnier loss $\mathcal{L}_{\text{Charb}}(x, y) = \sqrt{(x - y)^2 + \varepsilon^2}$ with $\varepsilon = 10^{-6}$ provides smoother gradients near zero compared to standard L1, which benefits fine-grained texture optimisation.

Multi-Scale Perceptual Loss

VGG-19 features are extracted at two layers for multi-scale supervision:

- **Stage 1** (`conv2_2`, layers 0–8): fine edges and low-level texture.
- **Stage 2** (`conv4_4`, layers 9–26): semantic structure and global context.

Both stages contribute equally ($w_1 = w_2 = 0.5$) to the perceptual L1 loss. The VGG-19 network is frozen throughout training.

FFT Frequency Loss

$$\mathcal{L}_{\text{FFT}}(x, y) = \| |\mathcal{F}(x)| - |\mathcal{F}(y)| \|_1$$

where $\mathcal{F}$ is the 2D real FFT (`torch.fft.rfft2`, `ortho` normalisation). This explicitly penalises high-frequency reconstruction errors that pixel and perceptual losses tend to under-penalise.

Adversarial Loss

Standard BCE with logits: $\mathcal{L}_{\text{adv}} = -E[\log D(G(x))]$. Label smoothing was applied to discriminator targets (real: 0.9, fake: 0.1) for training stability.

Composite Generator Objective (Phase 2)

$$\mathcal{L}_G = \underbrace{1.0 \cdot \mathcal{L}_{\text{Charb}}}_{\text{pixel}} + \underbrace{0.003 \cdot \mathcal{L}_{\text{Perc}}}_{\text{perceptual}} + \underbrace{0.1 \cdot \mathcal{L}_{\text{FFT}}}_{\text{frequency}} + \underbrace{0.0002 \cdot \mathcal{L}_{\text{adv}}}_{\text{adversarial}}$$

2.4. Training Strategy

Phase 1 — Pixel Pretraining (350 epochs)

Table 1: Phase 1 hyperparameters.

Parameter	**Value**
Optimiser	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4})
Initial LR	2×10^{-4}
LR Schedule	CosineAnnealingWarmRestarts ($T_0 = 50$, $T_{\text{mult}} = 2$)
Warm-restart cycles	3 (at epochs 50, 150, 350)
Loss	Charbonnier only
Gradient clipping	max_norm = 1.0
Mixed precision	AMP (`autocast` + `GradScaler`)
EMA decay	0.9995

Phase 2 — GAN Fine-tuning (50 epochs)

Table 2: Phase 2 hyperparameters.

Parameter	Value
Generator LR	1×10^{-5}
Discriminator LR	1×10^{-5}
LR Schedule (G & D)	CosineAnnealingLR ($T_{\max} = 50$, $\eta_{\min} = 10^{-7}$)
Label smoothing	real $\rightarrow$ 0.9, fake $\rightarrow$ 0.1
Generator loss	Charbonnier + Perceptual + FFT + Adversarial
Gradient clipping	max_norm = 1.0 (G and D separately)

2.5. Inference

EMA Generator

An Exponential Moving Average (EMA) of the generator weights was maintained throughout training with decay = 0.9995. At inference the EMA model consistently produced lower MAE than the standard checkpoint (see Section 3).

Test-Time Augmentation (TTA)

Self-ensemble over 8 geometric variants: all combinations of 4 rotations (0ř, 90ř, 180ř, 270ř) and horizontal flip (on/off). Each variant is independently predicted, the augmentation is reversed, and all 8 predictions are averaged pixel-wise. The notebook confirmed the output shape and range after TTA:

```
TTA output shape: torch.Size([1, 3, 128, 128]),
range: [0.013, 1.000]
```

3. Experiments and Results

3.1. Phase 1 Training Progression

The full epoch-by-epoch log from the notebook is summarised at representative checkpoints below. The Charbonnier loss and MAE both decrease consistently across all three warm-restart cycles, with each cycle restart temporarily raising the learning rate and allowing the model to escape local minima.

```
Epoch   1/350 | Charb Loss: 2.75837 | MAE(0-255):  90.98 | Best:  90.98 | LR:
     2.00e-04
Epoch  10/350 | Charb Loss: 0.07070 | MAE(0-255):  18.02 | Best:  18.02 | LR:
     1.81e-04
Epoch  20/350 | Charb Loss: 0.06897 | MAE(0-255):  17.58 | Best:  17.58 | LR:
     1.31e-04
Epoch  30/350 | Charb Loss: 0.06841 | MAE(0-255):  17.44 | Best:  17.44 | LR:
     6.92e-05
Epoch  40/350 | Charb Loss: 0.06806 | MAE(0-255):  17.35 | Best:  17.35 | LR:
     1.92e-05
```

```
Epoch  50/350 | Charb Loss: 0.06785 | MAE(0-255):  17.30 | Best:  17.30 | LR:
     2.00e-04  [restart]
Epoch 100/350 | Charb Loss: 0.06735 | MAE(0-255):  17.17 | Best:  17.15 | LR:
     1.00e-04
Epoch 150/350 | Charb Loss: 0.06677 | MAE(0-255):  17.03 | Best:  17.00 | LR:
     2.00e-04  [restart]
Epoch 200/350 | Charb Loss: 0.06702 | MAE(0-255):  17.09 | Best:  17.00 | LR:
     1.71e-04
Epoch 250/350 | Charb Loss: 0.06619 | MAE(0-255):  16.88 | Best:  16.87 | LR:
     1.00e-04
Epoch 300/350 | Charb Loss: 0.06555 | MAE(0-255):  16.71 | Best:  16.70 | LR:
     2.94e-05
Epoch 330/350 | Charb Loss: 0.06527 | MAE(0-255):  16.64 | Best:  16.63 | LR:
     4.99e-06
Epoch 350/350 | Charb Loss: 0.06527 | MAE(0-255):  16.64 | Best:  16.63 | LR:
     2.00e-04
  [Loaded best Phase 1 checkpoint (MAE: 16.63)]
```

Key observations: the model crossed the bicubic baseline (MAE 18.40) by epoch 10 and continued improving steadily. The third warm-restart cycle (epochs 150–350) provided the largest gains, pushing MAE from ≈17.00 to 16.63. The loss converged to approximately 0.0653 by the final epoch.

3.2. Phase 2 Training Progression

Phase 2 introduced the discriminator and the full composite loss. Discriminator loss stabilised from an initial 0.700 (epoch 1) down to ≈0.334 by epoch 30, indicating that generator and discriminator reached a stable equilibrium.

```
Epoch   1/50 | G Loss: 0.07340 | D Loss: 0.70020 | MAE(0-255): 16.68 | Best:
     16.68
Epoch  10/50 | G Loss: 0.07333 | D Loss: 0.36807 | MAE(0-255): 16.72 | Best:
     16.66
Epoch  20/50 | G Loss: 0.07213 | D Loss: 0.34061 | MAE(0-255): 16.68 | Best:
     16.66
Epoch  30/50 | G Loss: 0.07184 | D Loss: 0.33616 | MAE(0-255): 16.66 | Best:
     16.66
Epoch  40/50 | G Loss: 0.07190 | D Loss: 0.33423 | MAE(0-255): 16.68 | Best:
     16.66
Epoch  50/50 | G Loss: 0.07187 | D Loss: 0.33421 | MAE(0-255): 16.68 | Best:
     16.65
  [Loaded best Phase 2 checkpoint (MAE: 16.65)]
```

The conservative adversarial weight ($\lambda_{\text{adv}} = 0.0002$) ensured that the Phase 2 MAE remained within ≈0.02 of the Phase 1 best, protecting pixel accuracy while the discriminator added texture sharpness.

3.3. Model Comparison

The notebook output for the final model selection:

```
==================================================
```

Table 3: Model comparison on the full labeled training set (validation-style, no TTA).

Model / Method	**Evaluation**	**MAE (0–255)**
Bicubic interpolation	Full labeled pairs (1,642)	18.40
Phase 1 generator	Best training checkpoint	16.63
Phase 2 generator	Best training checkpoint	16.65
Standard generator	Full validation-style pass	16.6717
EMA generator	Full validation-style pass	**16.6675**

```
  Generator MAE (0-255):     16.6717
  EMA Generator MAE (0-255): 16.6675
==================================================
  -> Using EMA generator for inference (lower MAE)
```

3.4. Post-Training Analysis

A detailed per-image analysis was conducted on 200 randomly sampled validation images (no TTA) to understand the distribution of reconstruction quality:

```
============================================================
  POST-TRAINING ANALYSIS (200 samples, no TTA)
============================================================
  MAE  -- Mean: 16.95, Std: 6.19, Min: 5.29, Max: 31.56
  PSNR -- Mean: 21.69 dB, Std: 3.24 dB
============================================================
```

Table 4: Post-training per-image analysis on 200 validation samples (no TTA).

Metric	**Mean ± Std**	**Range**
MAE (0–255)	16.95 ± 6.19	5.29 – 31.56
PSNR (dB)	21.69 ± 3.24	—

The large standard deviation in MAE (±6.19) reflects variance across leaf types. The best single-image MAE of 5.29 corresponds to near-perfect reconstruction of uniform or low-frequency regions, while the worst case of 31.56 occurs on images with fine repeating vein structures where the degradation is most ambiguous.

4. Training Summary

The final notebook training summary, produced programmatically from the trained model objects:

```
============================================================
 TRAINING SUMMARY
============================================================


 Architecture:
   Generator: RRDB-Net with Channel Attention (SE blocks)
```

```
    - 14,792,003 parameters
    - 20 RRDB blocks, 64 base filters, 32 growth rate
    - ICNR-initialized PixelShuffle for 4x upsampling
  Discriminator: U-Net PatchGAN
    - 8,656,412 parameters
    - Per-pixel real/fake predictions with skip connections

 Training:
  Phase 1: 350 epochs, Charbonnier loss only
    - CosineAnnealingWarmRestarts (3 full cycles)
    - Best training MAE: 16.63
  Phase 2: 50 epochs, Charb + Perceptual + FFT + Adversarial
    - Very conservative (lam_adv=0.0002, lam_vgg=0.003)
    - Best training MAE: 16.65

 Inference:
  EMA generator (decay=0.9995) for smoother predictions
  8-fold Test-Time Augmentation (4 rotations x 2 flips)

 Key Design Decisions:
  Charbonnier loss > L1 for smoother gradients near zero
  FFT loss preserves high-frequency biological textures
  Multi-scale VGG perceptual loss (conv2_2 + conv4_4)
  AdamW optimizer with weight decay for regularization
  Best-model checkpointing prevents late-epoch regression
============================================================
```

5. Conclusion

The final solution used a 14.8M-parameter RRDB-Net generator with SE channel attention, trained in two phases on 1,642 plant leaf image pairs. Compared with the bicubic baseline MAE of 18.40, the model achieved a full-pass validation MAE of 16.67 — a relative improvement of $\approx 9.4\%$.

The most impactful design choices were:

1. **Charbonnier loss** for stable pixel-level convergence over 350 epochs.
2. **CosineAnnealingWarmRestarts** providing 3 full escape cycles that drove MAE from 18.02 (epoch 10) to 16.63 (epoch 330).
3. **FFT frequency loss** recovering high-frequency leaf texture detail that pixel and perceptual losses independently blurred.
4. **EMA generator** (decay 0.9995) consistently outperforming the standard checkpoint at inference.
5. **8-fold TTA** as a training-free performance boost at submission time.

A key limitation is that MAE was measured on the same labeled training distribution rather than a held-out split. Future work should reserve a stratified validation set for unbiased model selection and consider progressive resolution training to better utilise the limited 1,642-sample dataset.

6. References

1. Kaggle Plant Leaves Super-Resolution Challenge dataset (AgriVision).
2. X. Wang et al., *ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks*, ECCV Workshops, 2018.
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5. PyTorch documentation: pytorch.org/docs/stable/.
6. I. Loshchilov & F. Hutter, *Decoupled Weight Decay Regularization* (AdamW), ICLR, 2019.